# Project Initialization and Planning Phase

**Date:** 26 May 2026
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**Project Title:** Harnessing EC2, VPC & MobaXterm for Secure Web Hosting

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## Project Proposal (Proposed Solution)

This project demonstrates the deployment and hosting of a static web application using AWS cloud infrastructure. The solution leverages Amazon EC2 instances for hosting, VPC for secure network isolation, and MobaXterm for secure remote server management.

The main objective is to deploy a fully functional website on a public IP address while ensuring security, scalability, and accessibility. The deployed website is accessible via:

– **http://51.20.51.104/landing-page/#home**

The project showcases how cloud infrastructure can be used to host real-time web applications efficiently without relying on traditional hosting services.

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## Resource Requirements

| Resource Type | Description | Specification / Allocation |
|---|---|---|
| Hardware | Cloud Computing Resources | Amazon EC2 Virtual Machine (t2.micro / t3.micro / custom instance) |
| Network | Virtual Private Cloud (VPC) | Secure isolated network environment with public subnet |
| Access Tool | Remote Access Tool | MobaXterm SSH client for EC2 management |
| Storage | Instance Storage | EBS Volume (Elastic Block Storage) for website files |
| Software | Web Hosting Stack | Apache/Nginx Web Server |
| Development Tools | File Management | HTML, CSS, JavaScript for frontend development |
| Deployment Method | Hosting Approach | Static website deployment using EC2 public IP |
| Security | Access Control | Security Groups (SSH, HTTP enabled ports) |

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## System Architecture Overview

- User accesses website via public IP
- Request goes through AWS Internet Gateway
- Traffic routed to EC2 instance inside VPC

- Web server (Apache/Nginx) serves HTML/CSS/JS files
- MobaXterm used for secure SSH-based deployment and management

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## Project Description

The project is a real-time cloud deployment of a static website hosted on AWS EC2 instance. The application is deployed inside a secure VPC environment to ensure controlled access.

MobaXterm is used as a secure SSH client to connect to the EC2 instance, allowing file transfer, configuration, and server management.

The hosted website includes multiple pages such as:

- Landing page
- Home section
- Navigation menu
- Responsive UI design

The system ensures high availability and can be accessed globally using the public IP address assigned to the EC2 instance.

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## Data / Website Details

| Component | Description |
|---|---|
| Website Type | Static Web Application |
| Pages | HTML-based landing page |
| Styling | CSS for UI design |
| Functionality | JavaScript for interactivity |
| Hosting Platform | AWS EC2 |
| Access Method | Public IPv4 address |
| Deployment Tool | MobaXterm SSH |

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## Key Features

- Cloud-based website hosting using AWS EC2
- Secure SSH access using MobaXterm
- Publicly accessible web application
- Lightweight static website deployment
- Scalable cloud infrastructure

- VPC-based secure network environment
- Fast deployment and easy management

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## ⚡ Conclusion

This project successfully demonstrates cloud-based web hosting using AWS EC2 and VPC. It shows how a static website can be deployed and managed securely using MobaXterm, making it accessible worldwide through a public IP address.